

Auto-OPDx: Automated Profilometry Data Analysis for High-Throughput Experimentation

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Abstract—High-throughput combinatorial materials science platforms, such as Hypersurface Photolithography (HP), enable the rapid fabrication of patterned polymer brush arrays. Profilometry is a surface metrology technique that can be used to measure the topography, step height, and surface texture of these polymer brush arrays at the nanometer scale with high precision. However, proprietary software and manual interaction are needed to extract these features from the 3D scan data output by profilometers, usually with more than half an hour for a data array, which heavily bottleneck high-throughput experimentation. In this work, we tackle this bottleneck with Auto_OPDx, an open-source automated data processing pipeline designed for the rapid analysis of Bruker Dektak .OPDx profilometry data. This project implements a two-stage background correction process using global and local least-squares plane fitting and connected-component segmentation via OpenCV. Applied to fabricated samples from HP, Auto_OPDx achieves approximately 2 second processing times per 8x8 grid patterned substrate, eliminates operator bias, and reduces a critical bottleneck in calculating the height-to-condition correlation. Auto_OPDx is validated against experimenter reference measurements with $-0.039\ \mu\text{m}$ systematic bias. The code is posted on GitHub at https://github.com/cjeong1021/Auto_OPDx.

Index Terms—profilometry, automation, high-throughput characterization, image segmentation

I. INTRODUCTION

High-throughput experimentation (HTE) and self-driving laboratories (SDLs) have revolutionized materials discovery, enabling researchers to explore vast multi-dimensional design spaces at unprecedented speeds [1]. Modern robotic and lithographic techniques can fabricate hundreds of micrometer-scale features on a single substrate in minutes. Following fabrication, metrology tools such as a profilometer can quickly scan these substrates and produce 3D data for material characterization.

A primary example of this bottleneck occurs in the height and thickness characterization of micro-patterned arrays for transparent and conductive films using Hypersurface Photolithography (HP). In these systems, fabrication printing parameters (such as print time, UV exposure dose, or temperature) directly control the physical height of the printed features, which dictates the resulting material properties. For example,

a feature must be sufficiently thick to conduct electricity, yet thin enough to be transparent. If the feature height deviates from the optimal range, functional performance degrades. Therefore, mapping the feature heights is the key to correlating printing parameters with target material properties.

However, analyzing the output measurements remains a significant bottleneck due to the need for manual feature identification and selection. Currently, researchers must manually load 3D data into software, visually define regions of interest (ROIs) for each printed feature, and individually extract step heights. For a single substrate containing hundreds of features, this manual data processing requires hours of repetitive labor and introduces the risk of operator bias or fatigue.

To overcome this bottleneck, this paper introduces Auto_OPDx, a fully automated profilometry data extraction pipeline that is developed in Python. Auto_OPDx is designed to automatically load 3D data in bulk, detect patterned features using computer vision algorithms, and extract precise height measurements. Our contributions include:

- 1) By combining two-stage least-squares plane fitting with computer vision segmentation, an open-source, automated data processing pipeline – Auto_OPDx – is developed, successfully addressing the characterization bottleneck in high-throughput materials science.
- 2) The pipeline eliminates manual data analysis, with a 700 times reduction in processing time (from 30 minutes to 2.3 seconds per substrate), enables researchers to process profilometry data in real time.
- 3) The software was validated against 4,301 data points, demonstrating high accuracy ($R^2 = 0.9794$), allowing materials characterization to match automated fabrication techniques.

This paper is organized as follows. After the introduction in Section I, Section II discusses background and related work in profilometry. Section III presents the pipeline of Auto_OPDx, providing details of the methodology. Section IV describes experimental results. Section V concludes the paper and outlines future directions, including the development of a configurable open platform and extending the pipeline to process additional data types such as fluorescence.

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II. BACKGROUND AND RELATED WORK

A. Material Context: Transparent and Conductive Films

Transparent and conductive (T/C) films are essential components for modern electronic devices, including touchscreens, solar cells, and wearable sensors [2], [3]. A fundamental challenge in engineering these films is the inherent physical trade-off between electrical conductivity and optical transparency. Increasing the thickness of a conductive layer can improve conductivity by lowering the sheet resistance but it simultaneously blocks more visible light [4]. Recent advances in high-throughput materials science have introduced bottom-up patterning techniques, such as Hypersurface Photolithography (HP), to rapidly fabricate metal-polymer composite films [1]. In these systems, polymer brushes are grown directly on a substrate, and the physical height of these printed features dictates the volume of conductive material [1]. Because feature height is the primary factor deciding the final optoelectronic properties of the film, achieving fast and accurate height measurements across large sample sets is critical for optimizing the manufacturing parameters.

B. Profilometry in Material Characterization

Profilometers are the standard metrology tools used to measure surface topography, including step height, surface roughness, and film thickness. These systems generally fall into two categories: contact and optical [5].

1) *Contact Profilometry*: Contact tools operate by dragging a physical, often diamond-tipped, stylus across the surface of a sample (Fig. 1). As the probe moves, it physically traces the contours of the surface, generating a precise, two-dimensional topographical profile based on the stylus's vertical displacement. This method is highly established, deeply standardized across industries, and unaffected by the optical properties (such as reflectivity or transparency) of the measured material.

2) *Optical Profilometry*: Non-contact tools use light to map surface irregularities [5], [6]. Instead of a physical probe, these systems direct a light beam onto the sample and analyze the reflected interference patterns to capture a three-dimensional surface map. Optical profilometry would be more appropriate for non-rigid and delicate materials since there is no contact, but is problematic for transparent surfaces that do not reflect light.

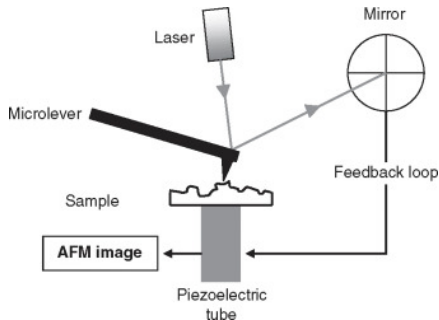


Fig. 1: Example of a contact profilometer that physically traces surface features [7] with a stylus.

A contact profilometer was used in this work, as the automated data processing pipeline was developed to assist in research for T/C films. Due to the transparency, optical profilometry would fail as it relies on measuring interference patterns from light reflected off the surface of the material.

III. METHODOLOGY

The data processing pipeline of Auto_OPDx is designed to automate the extraction of feature heights from profilometry scans of feature arrays. The pipeline consists of six primary stages (Fig. 2):

- 1) **Profilometry Data Loading**: .OPDx files are loaded and converted into NumPy coordinate and height matrices.
- 2) **Global Plane Estimation**: The global plane for the substrate is estimated using least-square approximation by thresholding 10px wide strips to pick background points.
- 3) **Feature Segmentation and Labeling**: Features are segmented from the background by thresholding against the estimated global plane.
- 4) **Connected Component Analysis**: Features are assigned labels and centroid locations using Connected Component Analysis.
- 5) **Local Plane Refinement**: A local plane is fit using least-square approximation using background points in a small area around each feature to minimize the effect of substrate warping.
- 6) **Height Extraction and Export**: Relative feature heights are computed, then compiled into a CSV file for materials characterization.

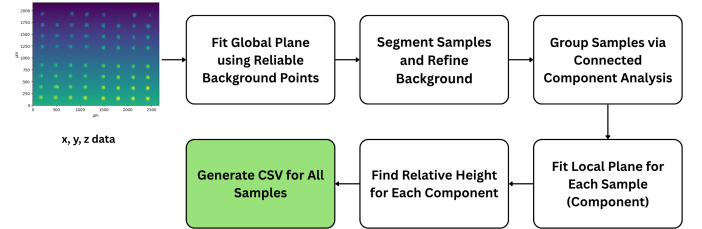


Fig. 2: Data processing pipeline for Auto_OPDx

A. Profilometry Data Loading

The measurement of patterned surfaces is performed using a Bruker Dektak stylus profilometer in a clean room to minimize dust and debris from interfering with data collection. An example of the sample surface containing arrays of printed features is shown in Fig. 3a.

The instrument outputs a proprietary .OPDx file containing precise surface measurements. To enable automated analysis, Auto_OPDx integrates the open-source OPDx_read library [8]. The data from the .OPDx file is parsed into three two-dimensional NumPy arrays representing the spatial coordinates and the height measurements:

$$x \in \mathbb{R}^{N_{cols}}, y \in \mathbb{R}^{M_{rows}}, z \in \mathbb{R}^{M_{rows} \times N_{cols}} \quad (1)$$

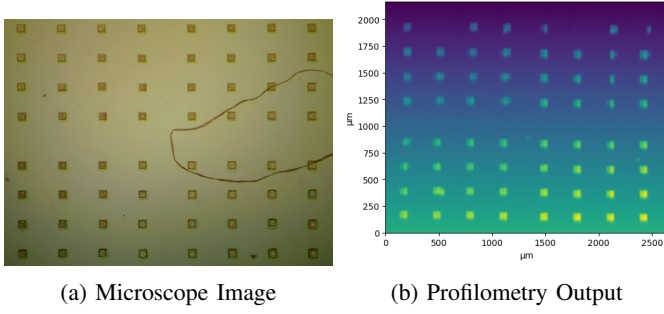


Fig. 3: Microscope image (left) showing the 8×8 grid array of features printed on a glass substrate and its 3D scan output (right) of profilometry .OPDx file.

where each element $z_{i,j}$ represents the measured surface height (in meters) at the coordinate (x_i, y_j) (in micrometers). The resulting height map is shown in Fig. 3b.

B. Global Plane Estimation

Often, mechanical tilt is introduced during sample mounting for stylus profilometers. This prevents a simple threshold from segmenting features from the background plane. To address this, the pipeline implements a stripwise thresholding approach to gather sufficient background points to estimate the background plane and correct the tilt.

The height matrix z is partitioned into K bands, each containing 10 rows in our experiments. For each band, background points are selected using an adaptive percentile threshold computed automatically via Otsu's method [9] on the height histograms. This stripwise calculation ensures that background heights are captured regardless of large-scale tilt and height variations from the features, as illustrated in Fig. 4. This percentile parameter can also be modified or configured by the user depending on sample coverage.

Thresholded Background Data with Fitted Plane

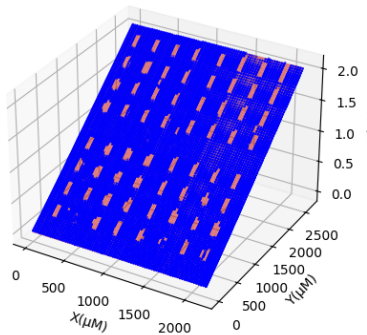


Fig. 4: Plot of global background plane estimation. The red plane represents the estimated global plane fitted to the blue colored background points.

A global linear plane is defined by the relation $z_{pred} = ax + by + c$. The coefficient vector $\mathbf{w} = [a, b, c]^T$ is solved

using ordinary least squares (OLS). The closed-form solution is given by:

$$\mathbf{w} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{z}_{BG} \quad (2)$$

where $\mathbf{A}$ is the matrix containing the coordinate rows $[x_{i,j}, y_{i,j}, 1]$ for all background pixels, and $\mathbf{z}_{BG}$ is the column vector of their corresponding heights. The global tilt is then subtracted to yield the corrected height map:

$$z_{corr,i,j} = z_{i,j} - (ax_{i,j} + by_{i,j} + c) \quad (3)$$

C. Feature Segmentation and Labeling

After leveling the data, a background mask $B_{i,j}$ is generated by identifying pixels within a stricter tolerance of the fitted global plane:

$$B_{i,j} = |z_{corr,i,j}| < h_{dist} \quad (4)$$

The foreground mask, representing the features, is defined as the logical negation $F_{i,j} = \neg B_{i,j}$, shown in Fig. 5.

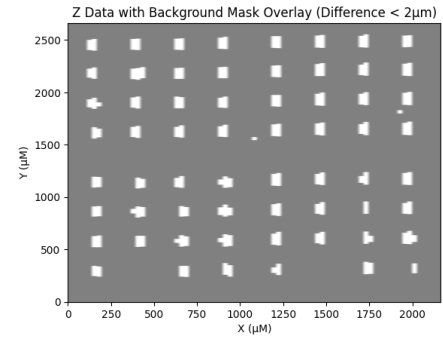


Fig. 5: Clean binary mask after applying the minimum distance threshold of $2\mu\text{m}$ from the global background plane, showing possible features.

Then Connected component analysis (CCA) is applied to the binary foreground image using OpenCV [10], [11] to group and label pixels into distinct features:

$$F_{i,j} \rightarrow \mathcal{L}_{i,j} \in \{0, 1, \dots, L\} \quad (5)$$

where $\mathcal{L}_{i,j} = 0$ is the background, and positive integers denote unique component labels. To remove noise and debris, components with an area A_l (in pixels) less than a minimum threshold A_{min} are discarded. The threshold A_{min} is adaptively set based on the median component area. Both the distance threshold h_{dist} and minimum area A_{min} can also be modified or fine-tuned by the user. For each valid component, the centroid coordinates $\mathbf{c}_l = (x_l^c, y_l^c)$ and bounding box dimensions around the centroids are calculated to define the regions of interest, as labeled in Fig. 6.

D. Local Plane Refinement

In addition to the mechanical tilt during sample mounting, thin glass or silicon substrates often exhibit localized warping that cannot be adequately corrected by a single global plane. To address this, a localized plane fitting is performed around each segmented component.

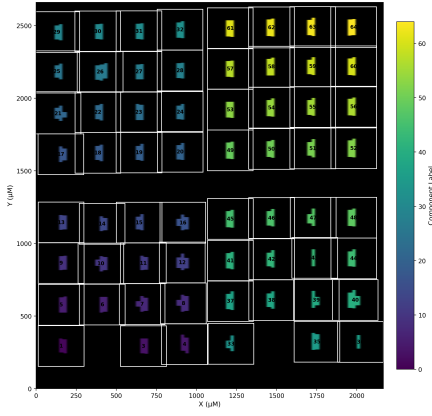


Fig. 6: Result of connected component analysis with bounding boxes around each feature.

For each labeled component l with centroid (x_l^c, y_l^c) , a localized bounding box is cropped around the centroid pixel by extending halfway to the adjacent components. The bounding box is larger than the feature to identify local background pixels using the refined background mask $B'_{i,j}$. If the local area contains sufficient background points ($N_{bg,local} \geq 3$), a local plane is fitted via OLS:

$$z_{pred,local} = a_{local}x + b_{local}y + c_{local} \quad (6)$$

This local fit models the specific surface slope directly beneath the individual feature, giving an accurate height estimate of the substrate underneath the feature that the profilometer cannot access. If the local region does not contain enough background points, the algorithm defaults to the global plane coefficients ($a_{local} = a$, $b_{local} = b$, $c_{local} = c$) to prevent singular matrix errors.

E. Height Calculation and Export

The physical height of the feature is calculated relative to the refined local reference plane. Within the local bounding box, the coordinates of the peak height are identified:

$$(x_{max}, y_{max}) = \arg \max_{(x,y) \in \text{Box}} z_{local}(x, y) \quad (7)$$

The predicted background level at the same coordinate is calculated using the local plane parameters:

$$z_{bg,max} = a_{local}x_{max} + b_{local}y_{max} + c_{local} \quad (8)$$

The relative height h_l of the feature (in micrometers) is computed as the difference between the peak height and this predicted background reference:

$$h_l = (\max(z_{local}) - z_{bg,max}) \times 10^6 \quad (9)$$

Finally, the relative heights, centroids, and reference Z-levels for all detected features are compiled and exported as a structured CSV file ready for material property characterization and optimization.

IV. RESULTS

The Auto_OPDx software was developed as a user-friendly package with a cross-platform Graphical User Interface, and was validated by analyzing 4,301 individual features across 69 distinct print runs from four separate substrate datasets. To assess the accuracy and runtime of the automated pipeline, all results were compared against measurements obtained by manually identifying regions of interest for each feature.

A. Graphical User Interface

To make the Auto_OPDx pipeline accessible to users without programming experience, a cross-platform Graphical User Interface (GUI) was developed using the PyQt5 framework. As shown in Fig. 7, the interface allows the selection of multiple files to process at the same time, with visualization of each .OPDx file and identified features on the right. The GUI also allows the user to fine tune parameters such as background percentile, minimum distance threshold and minimum area threshold.

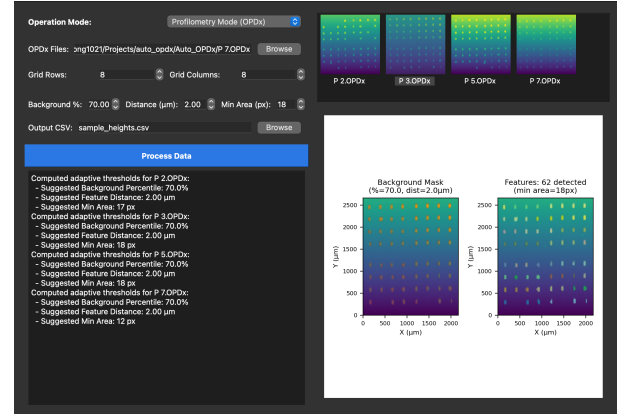


Fig. 7: Graphical User Interface with multi file selection and pre-processing visualization.

B. Measurement Consistency and Accuracy

The distribution of heights calculated by Auto_OPDx and those recorded manually is shown in Fig. 8. The automated results demonstrate excellent agreement with the manual records across all four datasets.

The pipeline achieved an overall Mean Absolute Error (MAE) of $0.0875 \mu\text{m}$ and a Root Mean Squared Error (RMSE) of $0.3749 \mu\text{m}$. The overall correlation coefficient (R) between the automated and manual measurements was 0.9897 ($R^2 = 0.9794$), indicating a strong linear agreement. The overall error (difference) distribution is shown in the histogram in Fig. 9. The summary statistics for each dataset category and the overall combined dataset are listed in Table I.

An analysis of the few cases with lower correlation revealed that the discrepancies were caused by high random noise and inconsistency in the manual spreadsheet records (where replicate measurements differed significantly from each other due to human error in placing the background markers). This

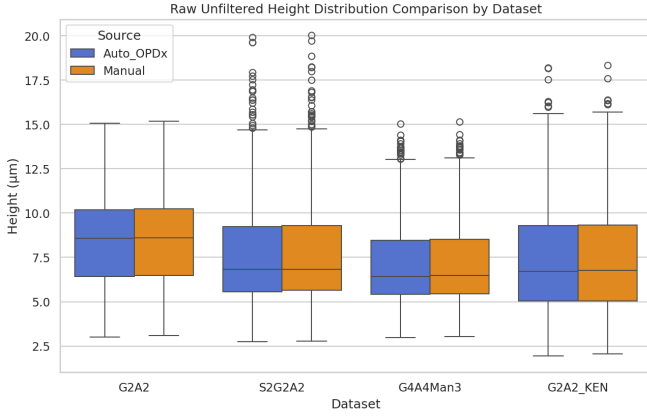


Fig. 8: Height distribution comparison between Auto_OPDx and manual spreadsheet records across the four evaluation datasets.

TABLE I: Auto_OPDx Summary Statistics on Four Substrate Datasets

Dataset	N	MAE (μm)	RMSE (μm)	R	R^2
G2A2	871	0.1014	0.2448	0.9803	0.9610
S2G2A2	986	0.1377	0.3317	0.9825	0.9653
G4A4Man3	2212	0.0603	0.0885	0.9986	0.9971
G2A2_KEN	232	0.1021	0.2947	0.9947	0.9893
Combined	4301	0.0875	0.3749	0.9897	0.9794

confirms that the automated pipeline provides a more consistent, deterministic, and objective measurement than manual analysis.

C. Throughput Resolution

The pipeline resolved the data analysis bottleneck by dramatically reducing processing times. For a standard substrate containing 64 features, manual analysis (loading the file, manually placing regions of interest, and saving results) required more than 30 minutes per substrate. Auto_OPDx

completed the entire pipeline in approximately **2.3 seconds** per file on a Macbook M1 Pro machine. This represents a throughput increase of over 700 times, enabling real-time data characterization.

V. CONCLUSION

This work successfully addresses the characterization bottleneck in high-throughput materials science by introducing Auto_OPDx, an open-source, automated data processing pipeline. By combining two-stage least-squares plane fitting with computer vision segmentation, the pipeline eliminates manual data analysis. The software was validated against 4,301 data points, demonstrating high accuracy ($R^2 = 0.9794$) and a 700 times reduction in processing time (from 30 minutes to 2.3 seconds per substrate). Auto_OPDx enables researchers to process profilometry data in real time, allowing materials characterization to match automated fabrication techniques. In the ongoing and future work, we would like to design an open platform with configuration interface that allows users to set the parameters we have used, and further process other kinds of data such as fluorescence data other than just profilometry data.

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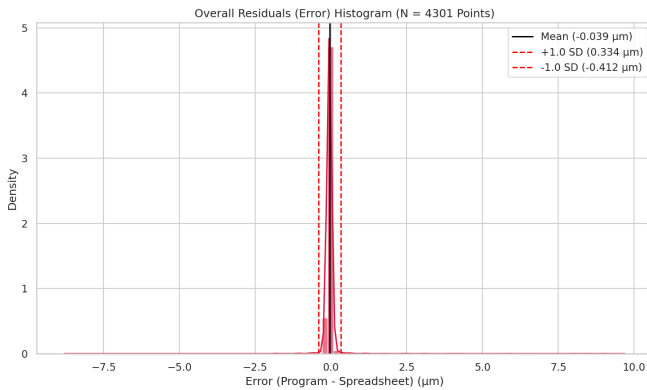


Fig. 9: Overall error histogram for 4,301 aligned data points, comparing Auto_OPDx results against manual records. The overall mean of $-0.039\mu\text{m}$ shows a small negative systematic bias.